

Valtti Järvi

FOUNDATIONAL TIME SERIES MODELS FOR EMERGENCY DEPARTMENT OCCUPANCY FORECASTING

Bachelor's Thesis
Faculty of Information Technology and Communication Sciences
Examiner: M.Sc. Eetu Pulkkinen
December 2025

ABSTRACT

Valtti Järvi: Foundational time series models for emergency department occupancy forecasting
Bachelor's Thesis
Tampere University
Bachelor's Programme in Computing and Electrical Engineering
December 2025

Emergency department overcrowding is a significant threat to patient safety..

Keywords: time series forecasting, foundational models, emergency department occupancy prediction

The originality of this thesis has been checked using the Turnitin OriginalityCheck service.

TIIVISTELMÄ

Valtti Järvi:
Kandidaatin tutkielma
Tampereen yliopisto
Tieto- ja sähkötekniikan kandidaattiohjelma
Joulukuu 2025

Avainsanat: keyword, keyword, keyword, keyword, keyword

Tämän julkaisun alkuperäisyys on tarkastettu Turnitin OriginalityCheck -ohjelmalla.

CONTENTS

1. Introduction	1
2. Background	3
2.1 Emergency department	3
2.1.1 Definition and consequences of crowding	3
2.1.2 Emergency department forecasting	4
2.2 Time series forecasting	4
2.3 Foundational models	5
2.3.1 Pre-training and fine-tuning paradigm	6
2.3.2 Selected foundational time series models	6
3. Methodology	7
3.1 Data	7
3.2 Experimental setup	8
3.3 Evaluation	9
3.4 Models	10
3.4.1 Chronos	10
3.4.2 TimesFM	11
3.4.3 Moirai	11
3.4.4 Sundial	12
3.4.5 TiRex	12
3.4.6 Benchmark models	12
4. Results and Discussion	14
4.1 Predictive performance	14
4.2 Effect of context length	15
4.3 Effect of covariates	16
4.4 Computational cost	17
4.5 Limitations	18
5. Conclusion	19
6. References	21
Appendix A: Computational costs for multivariate forecasts	25

1. INTRODUCTION

Emergency department (ED) crowding has been widely documented to negatively impact a range of clinically relevant patient outcomes [1], including higher mortality rates [2, 3] and longer lengths of stay [4]. Accurate predictions of future ED occupancy levels would enable proactive resource allocation aiming to alleviate or even prevent future crowding events.

Emergency department forecasting refers to the process of predicting the future status of the ED, typically using timeseries forecasting models. Traditionally, these have been trained using historical utilization data – sometimes with external variables such as weather and traffic data [5, 6]. However, training machine learning models involves a number of problems. First, every hospital has unique characteristics. A model trained with data from one hospital may not accurately predict occupancy levels for other hospitals. This means that a new model must be trained for each hospital at the time of implementation and repeatedly throughout the production cycle, which can be a computationally heavy and time-consuming task. Second, accessing the required data for each hospital can be a challenge due to either regulatory or practical reasons.

Foundation models (FMs) are artificial intelligence (AI) systems that can be used in a wide range of different tasks and are trained on massive amounts of diverse data [7]. In the context of emergency department forecasting, they offer a set of unique advantages in comparison to traditional machine learning algorithms. First, there is no need for model training. Second, they can be deployed with minimal local data collection. Third, their massive size and training data allows them to discover patterns that traditional machine learning model would have missed. [7]

Foundational models do not need to be trained on task-specific data. If they can predict ED occupancy accurately enough, the difficult task of model training can be skipped. This could save a substantial amount of computation and man-hours.

This thesis evaluates whether the newly introduced foundational time series models can be used to predict ED utilization of Tampere University Hospital ED "Acuta." A public dataset from a GitHub repository [8] containing the total hourly occupancy of the ED from June 20, 2018 to June 19, 2019 is used as the target variable, and the performance of the foundational models is evaluated in comparison to the state-of-the-art (SOTA) models

documented by [5].

The research questions of this thesis are as follows: (1) how does the performance of the FMs compare to the traditional SOTA models; 2) are there differences between the performances of the existing FM implementations, and 3) how do the FMs differ in terms of computational requirements?

The thesis is structured as follows. Chapter 2 provides the necessary background on emergency departments, crowding, time series forecasting, and foundational models. Chapter 3 details the experimental setup and the dataset used. Chapter 4 presents the empirical results and their implications in the context of existing knowledge. Finally, Chapter 5 provides final conclusions that can be drawn from the work.

2. BACKGROUND

(TARKISTA) This chapter provides the background for forecasting emergency department (ED) crowding. We first outline the role of EDs and clarify what crowding means and why it matters. We then introduce forecasting in the ED context and review core time series concepts used in this work. Finally, we present foundational time series models, explain the pre-training and fine-tuning paradigm, and describe the specific models evaluated in this study.

2.1 Emergency department

EDs deliver medical care to patients at any time with a wide range of acute conditions and injuries. [9]. They rapidly assess and treat urgent conditions. Swift diagnostic evaluation and timely medical or surgical interventions are provided for patients requiring immediate care [1].

2.1.1 Definition and consequences of crowding

Emergency departments (EDs) serve a critical role in healthcare systems by providing immediate access to medical services [1]. Worldwide, they function as a crucial 'safety net,' addressing acute health problems and responding to local and national disasters [10, 11]. In the United States, more than half of all inpatient hospital admissions originate from the ED [10].

ED crowding does not have a single widely accepted definition. The proposed frameworks include NEDOCS or EDWIN but their utilization and practicality is uncertain. Regardless of the definition, crowding is always

a situation in which demand for service exceeds the ability to provide care within a reasonable time, causing physicians and nurses to be unable to provide quality care

as was eloquently expressed by [12].

Regardless of the definition, crowding has become a worldwide public health issue (lähde), the negative impacts of which can be divided into three broad categories based on their

target: the patient, the staff, and the entire healthcare system. First, crowding has been repeatedly associated with increased mortality and increased number of medication errors [13](tarkista). Second, crowding has been shown to increase the stress levels of nurses and physicians [14]. Third, crowding results in ambulance diversions diverting ambulances to another ED [10], which delays patient access to care and disrupts coordination between hospitals. In addition, crowding compromises the ability of the ED to respond to mass emergencies or at least their capability to provide quality care to less urgent patients (lähde).

2.1.2 Emergency department forecasting

Emergency department forecasting refers to the process of predicting future utilization of the ED. Accurate forecasting would enable proactive resource allocation that could ultimately prevent overcrowding along with its harmful consequences. In previous literature, the predictability of a wide set of target variables has been investigated. These include the number of patient arrivals, occupancy, patient admission, length of stay, or occupancy ratio.[9]

Tähän kuuluu osio, jossa viitataan kunkin vastemuuttujan osalta tiettyyn juttuun (katso tuosta Gul et al). Sitten kerrotaan, että mitkä on kirjallisuuden puutteet. Lyhyimmillään: "The utilization of foundational models has previously only been reported by (viite

2.2 Time series forecasting

The goal of time series forecasting is to predict the future evolution of a data sequence based on its past values. External data sequences can be included for more accurate forecasts. [6] For example, future hourly ED occupancy can be predicted using past ED occupancy data, calendar information, and weather.

A discrete-time time series is a set of discrete values $\{x_t\}$, where t denotes a specific time [15]. In this thesis, the term 'time series' refers always to discrete-time time series. The intervals between t are usually fixed and of equal length.

The term univariate refers to data that consists of a single attribute or variable. In contrast, multivariate data consists of multiple related variables observed over time, where the relationships between variables can provide additional information for forecasting. It can be denoted as $\{\mathbf{x}_t\}$, where $\mathbf{x}_t$ is a vector. [15] Other variables than the one we predict are called covariates or exogenous inputs.

For a time series to be suitable for forecasting, it must exhibit identifiable patterns. The trend component represents the long-term direction or gradual change in the data, which, in the case of ED occupancy data, may reflect factors such as population growth. Seasonality

means repeating patterns that occur at regular intervals. For example, ED occupancy peaks daily between 16:00-19:00 and is at its lowest between 04:00-07:00 [5]. Random noise refers to irregular and random fluctuations that cannot be explained by any pattern. After all, many things, such as accidents happen randomly. These and many other patterns, including patterns from covariates, make forecasting possible. [15]

A *model* is a mathematical equation that describes the patterns found in the data. The objective is to find a model that effectively captures the underlying structure of the time series. This relationship can be formally expressed with a general equation:

$$y_{t+1} = f(y_t, y_{t-1}, \dots, \mathbf{x}_{t+1}, \mathbf{x}_t, \dots) + \epsilon_{t+1} \quad (2.1)$$

The function f represents the model itself. Specifies the relationship between the target variable y_{t+1} and the past values of the target variable $y_t, y_{t-1}, \dots$, and the present and past values of the covariates $\mathbf{x}_t, \mathbf{x}_{t-1}, \dots$. The term ϵ_t is the random noise that is not explained by the model. The function f can be a simple statistical model or a more complex machine learning model like a neural network. [6]

The future is always unpredictable to some extent. Therefore, predicting the entire probability distribution of a future value, rather than just a single value, is often useful. This is called a forecast distribution. Prediction intervals represents a range of values that the future value could take given some probability. For example, a 90 % prediction interval contains the actual value with 90 % probability. In contrast, a point forecast represents the best single estimate of a future value. It is often the mean or median of the forecast distribution. [6]

2.3 Foundational models

Foundational models are artificial intelligence (AI) systems that perform well in a wide range of diverse tasks. These models are trained on massive amounts of diverse data. Foundational models use deep neural network architectures with parameters ranging from hundreds of millions to hundreds of billions, depending on the application domain. [7] For example, foundation models have been developed for time series forecasting. A single foundation time series model can predict diverse time series data, from ED occupancy to stock prices.

Foundational models differ from traditional machine learning models that are trained for a specific purpose. For example, a model can be trained to forecast ED occupancy for one specific hospital. Advancements in computing have made it possible to train a larger model for multiple different purposes [7].

Transfer learning is a concept of transferring knowledge from one learned task to other tasks [16]. From vast training data, fundamental structures that are true for all tasks in one

domain, are learned [16]. For example, ED occupancy and stock prices could have same kind of daily fluctuation. Transfer learning and the vast size of foundation models is what makes them work [7].

In [7], the authors state three advancements that have enabled the size of foundation models. First, improvement in computational hardware. Second, the Transformer architecture that efficiently utilized this hardware for training [17]. Third, access to larger datasets for training.

2.3.1 Pre-training and fine-tuning paradigm

Deep learning is a part of machine learning that uses deep neural networks to learn from data. These networks learn data representations at several levels. [18] Deep learning powers most modern AI models. All foundation models in this thesis use deep learning, as does the TFT baseline [19].

In [20], the author outlines three key aspects of developing machine learning models. First step is choosing representation, or model. In case of deep learning, this is done by choosing a model architecture. Then the model is optimized or trained to fit the data. Training is done by adjusting the model parameters to minimize a loss function. This step is computationally intensive and often requires massive datacenters and large amount of time [7]. The final step is evaluation, where the model's performance is measured using data not seen during training. Deep learning follows this framework [18].

Foundational models use different framework and they represents a paradigm shift from traditional machine learning framework. For a specific task, already pre-trained foundation model can be selected. Foundation model is often fine-tuned for a specific task. After fine tuning, the model is evaluated. [7]

This approach saves a lot of computation, time and effort, since the foundation model is already pre-trained. It democratizes access to advanced AI models. In case of ED occupancy forecasting, every hospital has its own characteristics. Therefore model trained for one hospital would perform poorly for other hospitals. Training a separate model for each hospital would be expensive. Foundational models can solve this problem.

2.3.2 Selected foundational time series models

This section describes the foundational models selected for this study. The models are compared using their architecture and key forecasting parameters. The *context length* is the number of past time steps the model uses to make a prediction. The *forecasting horizon* is the number of future time steps it predicts. All selected models have promising zero-shot capabilities and can be fine-tuned for specific tasks.

3. METHODOLOGY

This section details the experiments used to evaluate whether foundation time series models are (good) for forecasting ED occupancy. A direct comparison with the baseline models from Tuominen et al. [5] requires the use of identical data and evaluation metrics.

First, the ED occupancy dataset and external covariates are described. Second, the experimental setup for 24-hour zero-shot forecasting task and different context lengths and external covariate setups are described. Used hardware and measurements for computational costs are also introduced. Third, the evaluation metrics are defined. Finally, all five foundation models and four benchmark models are described.

3.1 Data

ED occupancy was used as a forecasting metric by Tuominen et al. [5]. To directly compare results with Tuominen et al., this thesis uses ED occupancy as its crowding metric. All patients from all ED treatment spaces are summed up to this metric. All parts of the input-throughput-output model are affected in occupancy, which makes it useful. [5]

The ED occupancy data used in this thesis comes from Tampere University Hospital's (TAYS) electronic health record. TAYS is an academic, Level 1 trauma center in Tampere, Finland. When the study by Tuominen et al. [5] was done, TAYS served a primary population of 535,000 within the Pirkanmaa Hospital District and provided tertiary care for a total catchment population of 900,700. The hospital's combined emergency department (ED), Acuta, managed an annual census of approximately 100,000 patients at that time. The ED had a total capacity of 111–118 patients, which included 70 beds (plus 7 reserve) and 41 seats for ambulatory patients. [5]

The ED occupancy data spans from January 1, 2017, to June 19, 2019. It consists of hourly registered ED visits. As by Tuominen et al. [5], a whole year of occupancy data from June 20, 2018, to June 19, 2019, was used as a testing dataset in this thesis. Tuominen et al. also used a total of 167 external covariates. All of the covariate categories are described in table 3.1. Covariates can be divided into categories of past and future covariates. Future covariates include data points for the time steps for which the forecasts are made. For example, weather forecasts and public events can be used as future covariates. [5]

Category	Description
Hospital beds	Hourly measurements of available beds across 24 healthcare facilities within the catchment region.
Trends	Hourly Google Trends intensity for the keyword <i>Acuta</i> .
Traffic	Hourly vehicle throughput recorded at 33 bidirectional monitoring stations in the Pirkanmaa region.
Weather	Ten historical meteorological variables observed at the nearest weather station.
Public events	Daily counts of organized public events within the Tampere area.
Website visits	Web analytics for two hospital domains combined with Google search activity for "Acuta".
Calendar	Encodings of temporal attributes including weekday, month, national holidays, and associated lead-lag indicators.
Technical analysis (TA)	Autoregressive features derived from the target series, such as moving averages.
All categories (All)	Composite feature set of all 167 features comprising all categories.

Table 3.1. *Categories of External Covariates (adapted from Tuominen et al. [5])*

- where the data came from
- describe the data used (variables, time range,)
- how it was processed

3.2 Experimental setup

Models task is to predict the next 24 hours of hourly occupancy data. For a one year testing dataset, this means 365 predictions. All of the predictions are grouped together and compared against the ground truth values from testing dataset. These results are then directly compared with results from Tuominen et al. [5].

To evaluate univariate model performances across different historical window sizes, a set ten different context lengths were tested. These were grouped into temporal and fixed-token windows. The temporal contexts included calendar based intervals of 1 day, 3 days, 1 week, 2 weeks, 4 weeks, and 1 year. The maximum available window size from January 1, 2017 to June 19, 2018 spanned a total of 12840 hours or 535 days, was also

included. To fill the gaps in between these window sizes, fixed-length windows of 512, 1024 and 2048 hours were also evaluated.

To test which kind of external covariates foundation models can leverage, the covariates were divided into subsets according to their categories. To enable a direct comparison with Tuominen et al. [5], one forecasting was also performed with all covariates grouped together. Adding all the covariates dramatically increases the input dimensionality. This, in turn, rapidly scales the computational load especially with longer context lengths. This is why only context lengths 256, 512 and 1024 hours were tested.

Univariate forecasts were performed on a local workstation equipped with an Intel Core i5-8600K (6-core @ 4.3 GHz) CPU and 16 GB of DDR4 RAM. Multivariate forecasts were computed on the CSC Puhti supercomputer [**<empty citation>**]. These experiments were run on a GPU node with one 20-core Intel Xeon Gold 6230 CPU, and one NVIDIA Volta V100 GPU with 32 GB of VRAM.(tarkista tarkemmat tiedot)

For evaluation computational requirements of each foundation model, time needed to make forecasts and peak RAM during inference were measured. Time it takes to generate forecasts is called *inference time*. Inference time was measured for the whole test set spanning one year. Peak RAM usage is measured over the whole inference run, including loading the model to the computer memory.

- task is to predict 24h occupancy to the future (zero shot)
- details on different context length inputs, uni/multi
- hardware setup and library versions

3.3 Evaluation

In this thesis same metrics are used as in Tuominen et al. [5]. They used mean absolute error (MAE) and root mean squared error (RMSE) for point forecasts.

MAE is defined as

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t|, \quad (3.1)$$

where y_t is the ground truth value and $\hat{y}_t$ is the prediction made by the model. RMSE is defined as

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2}, \quad (3.2)$$

using the same notation.

MAE measures forecast accuracy in the same units as the data and treats all errors equally [6]. This makes it easy to interpret. A MAE of 5 means forecasts are off by 5 patients on

average.

RMSE squares the errors before averaging, which gives more weight to larger errors [6]. This makes RMSE more sensitive to outliers and particularly useful when large errors are more harmful than small ones. In ED occupancy forecasting, this property is valuable because large prediction errors could lead to understaffing or overstaffing. This could cause crowding or wasted resources.

3.4 Models

The models for this study were selected based on three criteria: open-source availability, ease of use, and performance on the GIFT-Eval benchmark [21]. All chosen models have promising zero-shot capabilities and are available for download from GitHub. Models were downloaded from GitHub on September 2025. Exact version of the cloned repository and model version name on Table 3.2. Median predictions were chosen over mean predictions for forecasts.

- settings and hyper-parameters that were used for each model (mean or median) -when and where downloaded
- exact version name for each
- short description of baseline models

3.4.1 Chronos

Chronos [22] is a family of foundational time series models developed by Amazon Science. It adopts a language modeling framework. This treats time series forecasting as a next-token prediction task. Based on the T5 transformer architecture [16], Chronos processes continuous time series data by scaling and quantizing it into a fixed vocabulary of discrete bins. The model is trained using cross-entropy loss to predict the next token. This frames forecasting as a classification problem at each time step. To generate forecasts, the model predicts one token at a time. Each predicted token is used as input for predicting the next one. These discrete tokens are then dequantized and inverse-scaled to produce continuous forecast values. Multiple forecast paths are sampled to compute quantile and mean forecasts.

The Chronos family includes four models with parameters ranging from 20 to 710 million. All variants have a maximum context length of 512 and a maximum forecasting horizon of 64. The training data was augmented using the TSMIXup and KernelSynth algorithms. TSMixup creates new time series by combining existing ones. KernelSynth generates synthetic data using Gaussian processes. As of September 2025, Exact version used called *chronos-bolt-base*, holds fourth place in GIFT-Eval for models with training data

overlap to the test set [23].

3.4.2 TimesFM

TimesFM, developed by Google Research [24], is a decoder-only transformer model. Its architecture consists of input layers, a transformer stack, and output layers. The model uses a technique called patching. Patching divides the input time series into fixed-length segments. The model processes these segments instead of individual time points. This reduces the number of steps required for long-horizon forecasting. Input patches are 32 time steps long. Output patches are 128 time steps long. The model is trained using Mean Squared Error (MSE) loss on point forecasts.

The latest version, TimesFM 2.5 that was used in this thesis was released in September 2025. It has 200 million parameters and context length of 16384. The model was pretrained on approximately 100 billion time points from diverse sources. These included Google Trends, Wikipedia(wikimedia)? pageview statistics, and public datasets such as M4 and Electricity. This data was augmented with synthetic data from ARMA processes. As of September 2025, TimesFM 2.5 holds third place on the GIFT-Eval leaderboard [23].

3.4.3 Moirai

Moirai, developed by Salesforce AI Research [25], is a versatile time series forecasting model. The version used in this thesis is Moirai 2.0 small. It was released on August 8, 2025 [26]. Moirai uses a novel decoder-only transformer architecture designed for heterogeneous multivariate time series. Its key feature is using varying patch sizes based on data frequency. Higher-frequency data uses larger patches. Each patch is processed by a linear layer and assigned a time ID and a variable ID. These IDs inject positional information into the transformer through rotary embeddings and learnable scalars. The transformer's output patches are projected to produce parameters for a mixture of probability distributions. These parameters include the weights for combining the distributions into a final weighted sum. This sum forms the final probabilistic forecast.

Version of Moirai that was used has approximately 12.4 million parameters. This is 96% smaller than the original Moirai Large model. Its architecture does not impose a hard limit on context length. It was trained on a combination of the GIFT-Eval dataset, synthetic data from KernelSynth and TSMIXup, and internal Salesforce data. As of September 2025, Moirai 2.0 small holds the sixth place on the GIFT-Eval leaderboard [23].

3.4.4 Sundial

Sundial is a family of models developed at Tsinghua University [27]. It was released on February 2, 2025. It is the first foundational time series model to output probabilistic predictions directly. This is done by producing multiple forecast samples, where mean median and quantile forecasts are calculated. Sundial uses an encoder-only transformer architecture and works directly on continuous values. The model processes time series data in patches of size 16. It is trained using a novel TimeFlow Loss function. A key feature is its non-autoregressive prediction mechanism. It predicts multiple patches at once, enabling millisecond-fast inference.

Version of the Sundial that was used has a context length of 2880 but can handle shorter, arbitrary context lengths. The model was trained on TimeBench. This is a large-scale dataset created by the authors, which contains over a trillion data points from real-world sources like weather and finance. As of November 2025, Sundial holds the 10th place on the GIFT-Eval leaderboard [23]. This demonstrates strong zero-shot forecasting capabilities.

3.4.5 TiRex

TiRex was developed by NXAI GmbH in collaboration with academic researchers. It was released on May 29, 2025 [28]. It is the only model in this thesis based on the xLSTM architecture rather than a transformer [29]. TiRex normalizes inputs using z-score normalization and processes data in patches of size 32. Input patches are projected into an embedding dimension by a residual block. This embedding is fed into the xLSTM. A second residual block projects the xLSTM output back into the patch dimension to produce forecasts. The outputs are scaled back to the original target space. The model predicts nine quantiles for each time step.

TiRex has 35 million parameters, making it computationally efficient. Its xLSTM architecture propagates hidden states from patch to patch. This removes hard constraints on context length. A length of 2048 was used in the original paper. The model’s training data comprises 47.5 million time series from Chronos’s training data, synthetic data similar to KernelSynth, and parts of the GiftEval pre-training dataset. As of November 2025, TiRex holds second place on the GIFT-Eval leaderboard among models with training data overlap to the test set [23].

3.4.6 Benchmark models

The Seasonal Naïve (SNaïve) model provides a baseline forecast by repeating the last observed value from the previous seasonal period. For a time series with a daily pattern,

Model	Version Name	Repository
Model 1	chronos-bolt-base	<code>\cite{model1onpaper}/model-1-long-name-that-needs</code>
Model 2	timesfm-2.5-200m-pytorch	<code>\cite{model2onpaper}/model-2</code>
Model 3	moirai-2.0-R-small	<code>\cite{model3onpaper}/time-series-foundation/lag-llama</code>
Model 4	TiRex	<code>\cite{model4onpaper}/model-4</code>
Model 5	sundial-base-128m	<code>\cite{model5onpaper}/model-5</code>

Table 3.2. Open-source models evaluated in this study.

the forecast for a future time is simply the value observed at the same time one seasonal cycle ago (e.g., 24 hours prior). This approach is a standard benchmark for strongly seasonal data [6].

ARIMAX (Autoregressive Integrated Moving Average with eXogenous inputs) is a statistical model that extends the standard ARIMA method. It combines ARIMA, which models the time series's own autocorrelation structure, with a linear regression component. This regression component allows the model to incorporate the predictive information from external, or exogenous, variables [6].

LightGBM (Light Gradient Boosting Machine) is a machine learning algorithm based on gradient boosting decision trees [30]. It is recognized for its high efficiency and scalability, which makes it particularly effective for large-scale datasets. Tuominen et al. [5] applied LightGBM in their study forecasting ED occupancy, where it achieved the strongest performance among the models evaluated.

The Temporal Fusion Transformer (TFT) is a deep neural network architecture designed specifically for multi-horizon time series forecasting [19]. It utilizes an attention-based encoder-decoder structure to learn complex temporal relationships at different scales. A key feature of TFT is its integrated interpretability, which allows for identifying significant past time steps and important features. This model was also benchmarked in the ED occupancy forecasting study by Tuominen et al. [5].

4. RESULTS AND DISCUSSION

In this thesis, we establish x main findings regarding the utilization of FM models in predicting ED occupancy. Each of these along with their implications are discussed below.

4.1 Predictive performance

For each foundation model, the result from best performing context length setting was picked. For TimesFM, TiRex and Sundial it was the largest context length that was tested. For TimesFM and TiRex it was 12840 hours and for Sundial 2048 hours. Moirai got its best result for second largest context length that was 8760 hours or one year. For Chronos, best result came using context length of 336 hours or two weeks. Best result using external covariates come with context length of 1024.

Table 4.1 shows that all univariate models except Chronos moderately outperformed best baseline model, LightGBM. Univariate models benefited from the large context length. Reason for poor performance of Chronos could be explained by its context length limit of 512 timesteps. All the other models supported much larger context lengths, allowing them to use the long range dependencies of the data. TimesFM outperformed other foundation models.

Model	FS	MAE	RMSE
Moirai	A	11.10	13.69
SN	U	9.53	12.58
ARIMA	U	7.78	10.50
Chronos	U	7.25	9.81
TFT	U	7.18	9.58
TFT	A	6.86	9.04
LightGBM	U	6.66	8.95
LightGBM	A	6.63	8.77
Sundial	U	6.57	8.85
TiRex	U	6.51	8.78
Moirai	U	6.35	8.59
TimesFM	U	<u>6.29</u>	<u>8.45</u>

Table 4.1. Best MAE and RMSE for each model. For each model, result from the best context length was picked. Foundational models are denoted with bold font. FS denotes Feature Set, where U is univariable and A is all covariates.

4.2 Effect of context length

No significant differences in performance were observed between the FMs when using the same context length. Performance keeps getting better fairly consistently when context length is increased. However, performance gains begin to diminish at around context lengths beyond 2048 hours. Table 4.1

In section 3.4 content of training data for each model is introduced. Every models training data consist of massive amounts time series data. All the data is drawn from the same finite pool of public and well-known time series data. Fairly similar results could be explained by this overlap of model training data. A second possible explanation is that the models have captured all the patterns within the historical data. Remaining residual component consists of random noise.

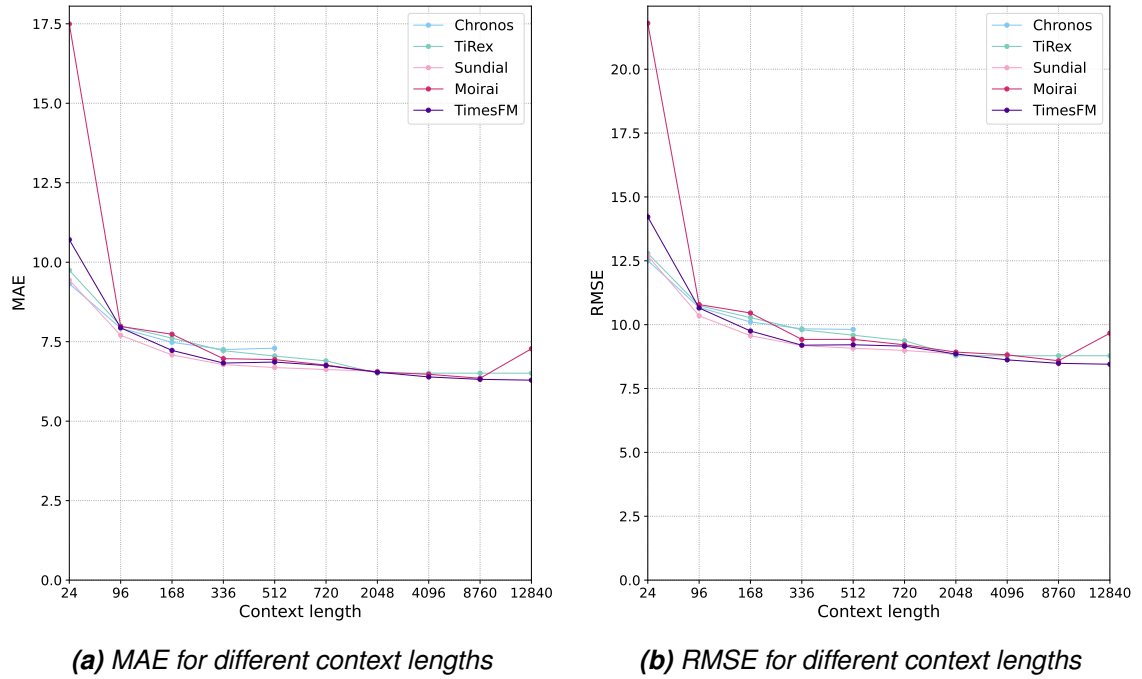


Figure 4.1. MAE and RMSE for different context lengths. All models take univariate input.

4.3 Effect of covariates

Moirai was the only model tested in this thesis which supported external covariates. Table 4.2 shows that surprisingly Moirai could not improve its performance with external covariates. Its performance got a lot worse.

This outcome suggests several possibilities. First, the chosen covariates may have been irrelevant and mostly random noise with no real correlation with the target variable. Second, the model inherently does not support covariates like this. Third, there could be a bug.

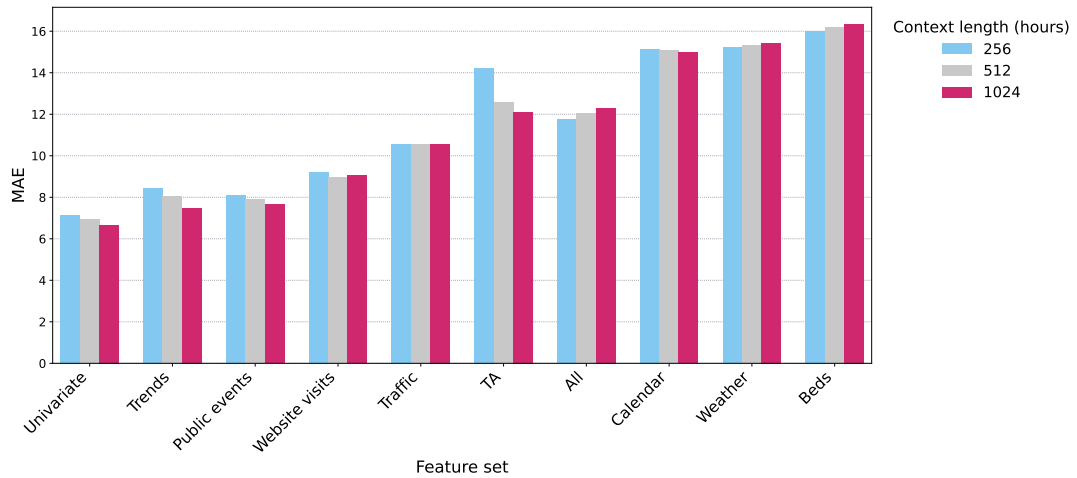


Figure 4.2. MAE across different multivariate feature sets. The "All" category includes all covariates, while other groups represent specific covariate subsets. Different colors show results for different context lengths.

4.4 Computational cost

Table 4.2 shows the computing time (latency) in seconds for each forecasting model across different context lengths. Table 4.3 presents the peak RAM usage in GiB for each model. Missing values indicate the model does not support that context length.

Table 4.3 shows that Moirai is the most memory efficient model. Table 4.2 shows that for small context lengths Chronos is the fastest model and for larger context lengths TiRex and Moirai are the fastest.

Context Length	Chronos	Sundial	TiRex	TimesFM	Moirai	Moirai (all)
24	1.5	173	104	48.2	6.5	—
96	2.4	162	107	63.9	5.2	—
168	4.0	170	91.8	65.6	5.6	—
336	6.7	169	92.9	68.8	5.8	—
512	9.5	172	94.0	70.8	7.2	110
720	—	189	91.5	73.9	12.3	—
2048	—	192	94.3	134	17.3	—
4096	—	—	103	197	20.6	—
8760	—	—	93.5	437	70.8	—
12840	—	—	98.5	635	94.6	—

Table 4.2. Computing time (latency) in seconds for each forecasting model across different context lengths. Latency represents the total time required to generate predictions for the entire test set spanning one year. Missing values indicate the model does not support that context length. Note that multivariate predictions were made on more powerful hardware.

Context Length	Chronos	Sundial	TiRex	TimesFM	Moirai	Moirai (all)
24	1.73	2.89	1.58	2.51	0.78	—
96	1.87	3.18	1.59	2.51	0.78	—
168	2.05	3.45	1.59	2.51	0.78	—
336	2.51	3.99	1.59	2.51	0.78	—
512	2.59	3.65	1.59	2.51	0.78	14.0
720	—	3.99	1.59	2.51	0.78	—
2048	—	6.79	1.59	2.51	0.78	—
4096	—	—	1.60	2.51	0.78	—
8760	—	—	1.62	2.51	0.78	—
12840	—	—	1.64	2.51	0.78	—

Table 4.3. Peak RAM usage in GiB for each forecasting model across different context lengths. Peak RAM represents the maximum memory consumed throughout the entire forecasting pipeline, including data preprocessing, model loading, and prediction generation. Missing values indicate that model does not support this length. Note that multivariate predictions from Moirai were done on different hardware.

4.5 Limitations

-only one dataset

5. CONCLUSION

Emergency department (ED) crowding worsens clinical outcomes and strains staff and resources. This thesis examined whether recently released foundational time series models (FMs) can forecast hourly ED occupancy accurately enough in a zero-shot setting to be useful in practice. Using Tampere University Hospital’s ED (Acuta) data and the experimental protocol aligned with prior work, we compared five open-source FMs against established baselines and assessed how context length, external covariates, and computational requirements affect their utility.

First, we evaluated predictive accuracy. All univariate FMs except Chronos moderately outperformed the strongest baseline (LightGBM). TimesFM achieved the best overall accuracy, while TiRex, Sundial, and Moirai were close when given similarly long histories. Chronos underperformed, likely due to its shorter maximum context. Increasing context length consistently improved accuracy across models, with diminishing returns beyond roughly 2,048 hours. These results show that zero-shot FMs can deliver competitive—and in most cases superior—accuracy for ED occupancy forecasting without any site-specific training.

Finally, we measured computational costs. Moirai was the most memory-efficient model. Chronos produced the fastest inferences at short contexts, while TiRex and Moirai were fastest at longer contexts. Inference times for all models were practical for hourly operational use on modern hardware. These findings indicate that zero-shot FMs can reduce the operational burden associated with frequent model retraining, enabling faster experimentation and easier deployment across sites.

This work has limitations. We studied a single hospital over a fixed period and evaluated one FM with multivariate inputs. We focused on zero-shot use; fine-tuning and lightweight adaptation were out of scope. We did not evaluate calibration, robustness to distribution shift, or the downstream operational impact of forecast-driven interventions. Future research should (i) replicate across multiple hospitals and time periods; (ii) test adapter-based fine-tuning and prompt conditioning; (iii) investigate covariate selection, forecasting, and representation strategies; (iv) evaluate probabilistic calibration and reliability under shift; and (v) run prospective, real-time studies measuring staffing, patient flow, and safety outcomes.

In conclusion, foundational time series models already provide accurate, fast, and operationally attractive zero-shot forecasts for ED occupancy. With long historical context, several models cluster in performance and surpass strong baselines, substantially lowering the barrier to adoption by removing per-site training. While multivariate zero-shot use did not help in our setting, careful covariate design and light adaptation may close that gap. Taken together, these results support zero-shot FMs as a practical starting point for ED forecasting and outline a path toward even stronger, fine-tuned, and reliably calibrated deployments.

6. REFERENCES

- [1] Sartini, M., Carbone, A., Demartini, A., Giribone, L., Oliva, M., Spagnolo, A. M., Cremonesi, P., Canale, F. and Cristina, M. L. Overcrowding in Emergency Department: Causes, Consequences, and Solutions—A Narrative Review. *Healthcare* 10.9 (Aug. 25, 2022), p. 1625. ISSN: 2227-9032. DOI: 10.3390/healthcare10091625. (Visited on 09/25/2025).
- [2] Berg, L. M., Ehrenberg, A., Florin, J., Östergren, J., Discacciati, A. and Göransson, K. E. Associations Between Crowding and Ten-Day Mortality Among Patients Allocated Lower Triage Acuity Levels Without Need of Acute Hospital Care on Departure From the Emergency Department. *Annals of Emergency Medicine* 74.3 (Sept. 1, 2019), pp. 345–356. ISSN: 0196-0644. DOI: 10.1016/j.annemergmed.2019.04.012. URL: <https://www.sciencedirect.com/science/article/pii/S0196064419303312> (visited on 09/25/2025).
- [3] Jo, S., Jin, Y. H., Lee, J. B., Jeong, T., Yoon, J. and Park, B. Emergency Department Occupancy Ratio is Associated With Increased Early Mortality. *Journal of Emergency Medicine* 46.2 (Feb. 1, 2014), pp. 241–249. ISSN: 0736-4679. DOI: 10.1016/j.jemermed.2013.05.026.
- [4] McCarthy, M. L., Zeger, S. L., Ding, R., Levin, S. R., Desmond, J. S., Lee, J. and Aronsky, D. Crowding Delays Treatment and Lengthens Emergency Department Length of Stay, Even Among High-Acuity Patients. *Annals of Emergency Medicine* 54.4 (Oct. 1, 2009), 492–503.e4. ISSN: 0196-0644. DOI: 10.1016/j.annemergmed.2009.03.006.
- [5] Tuominen, J., Pulkkinen, E., Peltonen, J., Kanninen, J., Oksala, N., Palomäki, A. and Roine, A. Forecasting emergency department occupancy with advanced machine learning models and multivariable input. *International Journal of Forecasting* 40.4 (Oct. 1, 2024), pp. 1410–1420. ISSN: 0169-2070. DOI: 10.1016/j.ijforecast.2023.12.002. URL: <https://www.sciencedirect.com/science/article/pii/S0169207023001346> (visited on 09/11/2025).
- [6] Hyndman, R. J. and Athanasopoulos, G. *Forecasting: Principles and Practice*. 2nd ed. Accessed on <current date>. Melbourne, Australia: OTexts, 2018. URL: <https://otexts.com/fpp2/>.
- [7] Bommasani, R. et al. *On the Opportunities and Risks of Foundation Models*. 2022. arXiv: 2108.07258 [cs.LG]. URL: <https://arxiv.org/abs/2108.07258>.

- [8] Tuominen, J., Pulkkinen, E., Peltonen, J., Kanninen, J., Oksala, N., Palomäki, A. and Roine, A. *ed-ml-multivar: Forecasting Emergency Department Occupancy with Advanced Machine Learning Models and Multivariable Input*. <https://github.com/tuomijal/ed-ml-multivar>. Code and data for: <https://doi.org/10.1016/j.ijforecast.2023.12.002>. 2023.
- [9] Gul, M. and Celik, E. An exhaustive review and analysis on applications of statistical forecasting in hospital emergency departments. *Health Systems* 9.4 (2018). No potential conflict of interest was reported by the authors., pp. 263–284. ISSN: 2047-6965. DOI: 10.1080/20476965.2018.1547348. URL: <https://www.tandfonline.com/doi/full/10.1080/20476965.2018.1547348>.
- [10] Asplin, B. R., Magid, D. J., Rhodes, K. V., Solberg, L. I., Lurie, N. and Camargo Jr, C. A. A conceptual model of emergency department crowding. *Annals of emergency medicine* 42.2 (2003), pp. 173–180.
- [11] Richardson, D. B. Increase in patient mortality at 10 days associated with emergency department overcrowding. *The Medical Journal of Australia* 184.5 (Mar. 6, 2006), pp. 213–216. ISSN: 0025-729X. DOI: 10.5694/j.1326-5377.2006.tb00204.x. URL: <https://www.mja.com.au/journal/2006/184/5/increase-patient-mortality-10-days-associated-emergency-department-overcrowding>.
- [12] : *Canadian Journal of Emergency Medicine* 3.2 (2001), pp. 82–84. DOI: 10.1017/S1481803500005285.
- [13] Kulstad, E. B., Sikka, R., Sweis, R. T., Kelley, K. M. and Rzechula, K. H. ED overcrowding is associated with an increased frequency of medication errors. *The American Journal of Emergency Medicine* 28.3 (2010), pp. 304–309. ISSN: 0735-6757. DOI: <https://doi.org/10.1016/j.ajem.2008.12.014>. URL: <https://www.sciencedirect.com/science/article/pii/S0735675708008310>.
- [14] Bond, K., Ospina, M., Blitz, S., Afilalo, M., Sam, G., Bullard, M., Innes, G., Holroyd, B., Curry, G., Schull, M. and Rowe, B. Frequency, Determinants and Impact of Overcrowding in Emergency Departments in Canada: A National Survey. *Healthcare quarterly (Toronto, Ont.)* 10 (Feb. 2007), pp. 32–40. DOI: 10.12927/hcq.2007.19312.
- [15] Peter J. Brockwell, R. A. D. *Introduction to Time Series and Forecasting*. 2016. DOI: 10.1007/978-3-319-29854-2.
- [16] Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., Zhou, Y., Li, W. and Liu, P. J. Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer. *The Journal of Machine Learning Research* 21.140 (2020), pp. 1–67. URL: <http://jmlr.org/papers/v21/20-074.html>.
- [17] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L. and Polosukhin, I. Attention Is All You Need. *Advances in Neural Information Processing Systems*. 2017, pp. 5998–6008. URL: <https://arxiv.org/abs/1706.03762>.

- [18] LeCun, Y., Bengio, Y. and Hinton, G. Deep Learning. *Nature* 521.7553 (2015). Published online: 27 May 2015; Issue date: 28 May 2015, pp. 436–444. DOI: 10.1038/nature14539. URL: <https://www.nature.com/articles/nature14539>.
- [19] Lim, B., Arik, S. Ö., Loeff, N. and Pfister, T. Temporal Fusion Transformer for Interpretable Multi-horizon Time Series Forecasting. *arXiv preprint arXiv:1912.09363* (2021).
- [20] Domingos, P. A Few Useful Things to Know About Machine Learning. *Communications of the ACM* 55.10 (2012), pp. 78–87. DOI: 10.1145/2347736.2347755.
- [21] Aksu, T., Woo, G., Liu, J., Liu, X., Liu, C., Savarese, S., Xiong, C. and Sahoo, D. GIFT-Eval: A Benchmark For General Time Series Forecasting Model Evaluation. *arXiv preprint arXiv:2410.10393* (2024).
- [22] Ansari, A. F., Stella, L., Turkmen, C., Zhang, X., Mercado, P., Shen, H., Shchur, O., Rangapuram, S. S., Arango, S. P., Kapoor, S., Zschiegner, J., Maddix, D. C., Wang, H., Mahoney, M. W., Torkkola, K., Wilson, A. G., Bohlke-Schneider, M. and Wang, Y. *Chronos: Learning the Language of Time Series*. Nov. 4, 2024. DOI: 10.48550/arXiv.2403.07815. arXiv: 2403.07815. URL: <http://arxiv.org/abs/2403.07815> (visited on 09/11/2025).
- [23] Salesforce AI Research. *GIFT-Eval Leaderboard: General Time Series Forecasting Model Evaluation*. <https://huggingface.co/spaces/Salesforce/GIFT-Eval>. Accessed: October 7, 2025. 2024.
- [24] Das, A., Kong, W., Sen, R. and Zhou, Y. *A decoder-only foundation model for time-series forecasting*. Apr. 17, 2024. DOI: 10.48550/arXiv.2310.10688. arXiv: 2310.10688. (Visited on 09/11/2025).
- [25] Woo, G., Liu, C., Kumar, A., Xiong, C., Savarese, S. and Sahoo, D. *Unified Training of Universal Time Series Forecasting Transformers*. May 22, 2024. DOI: 10.48550/arXiv.2402.02592. arXiv: 2402.02592. URL: <http://arxiv.org/abs/2402.02592> (visited on 09/11/2025).
- [26] Salesforce AI Research. *Introducing Moirai 2.0*. <https://www.salesforce.com/blog/moirai-2-0/>. Accessed: October 7, 2025. 2025.
- [27] Liu, Y., Qin, G., Shi, Z., Chen, Z., Yang, C., Huang, X., Wang, J. and Long, M. *Sundial: A Family of Highly Capable Time Series Foundation Models*. May 30, 2025. DOI: 10.48550/arXiv.2502.00816. arXiv: 2502.00816. URL: <http://arxiv.org/abs/2502.00816> (visited on 09/11/2025).
- [28] Auer, A., Podest, P., Klotz, D., Böck, S., Klambauer, G. and Hochreiter, S. *TiReX: Zero-Shot Forecasting Across Long and Short Horizons with Enhanced In-Context Learning*. May 29, 2025. DOI: 10.48550/arXiv.2505.23719. arXiv: 2505.23719. URL: <http://arxiv.org/abs/2505.23719> (visited on 09/11/2025).
- [29] Beck, M., Pöppel, K., Spanring, M., Auer, A., Prudnikova, O., Kopp, M. K., Klambauer, G., Brandstetter, J. and Hochreiter, S. xLSTM: Extended Long Short-Term Memory.

The Thirty-eighth Annual Conference on Neural Information Processing Systems.
Nov. 2024.

- [30] Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q. and Liu, T.-Y. LightGBM: A Highly Efficient Gradient Boosting Decision Tree. *Advances in Neural Information Processing Systems*. Vol. 30. 2017, pp. 3146–3154. URL: <https://proceedings.neurips.cc/paper/2017/file/6449f44a102fde848669bdd9eb6b76fa-Paper.pdf>.

APPENDIX A: COMPUTATIONAL COSTS FOR
MULTIVARIATE FORECASTS

Feature Subset	Context Length		
	256	512	1024
All	43.3	110	370
Beds	5.3	5.6	9.9
Calendar	5.3	5.4	5.6
Events	5.2	5.2	5.4
TA	5.5	7.8	15.6
Traffic	7.4	14.1	39.2
Trends	5.1	5.2	5.3
Weather	7.3	7.5	8.2
Website	5.2	5.2	5.4

Table A.1. Computing time (latency) in seconds for different feature subsets across context lengths (Moirai model). "All" represents all covariates combined, while other columns represent individual covariate features.

Feature Subset	Context Length		
	256	512	1024
All	1030	3730	14300
Beds	68.4	108	252
Calendar	56.5	66.1	90.0
Events	53.5	53.8	54.9
TA	84.4	171	498
Traffic	146	400	1400
Trends	53.5	54.3	55.3
Weather	62.4	78.9	144
Website	53.9	55.8	64.9

Table A.2. Peak RAM usage in MiB for different feature subsets across context lengths for Moirai model. "All" represents all covariates combined, while other columns represent covariate subsets.